MELODIUS TWIN PIANO

Play Together.
Feel Together.

Connecting hearts across miles through the magic of music

Feel Every Note, No Matter the Distance

https://youtu.be/LaLsGw5UT-Q

MEET THE TEAM

The Minds Behind Melodius

Lizi Wang

Hardware Software Developer

Circuit design, ESP32 programming, real-time audio synthesis, LED interaction, and MQTT communication

Ananya Jagadeesh Kedlaya

Software & Connectivity Lead

WiFi protocol design, real-time sync & latency optimisation

Xintong Shao

Industrial Design Lead

Enclosure design and Aesthetic

Madina Diallo

Project Manager & Research

Project coordination, User research, documentation & presentation strategy

THE CHALLENGE WE SOLVE

Music Connects Us - Distance Shouldn't Stop That

THE PROBLEM

- Distance removes **shared physical experiences**, not just proximity
- Video calls create communication , but not **presence**
- There is no way to **physically interact through music across distance**
- Existing digital instruments offer no physical mirroring ; if you press a key, nothing happens at the other end
- Users and learners miss the spontaneous, tactile connection of a shared instrument

THE SOLUTION

Melodius Twin Piano

Two identical pianos in different locations, connected over Wi-Fi. When you press a key, the same key moves on the other piano, in real time.
Not a video. Not a simulation. A real physical connection

WHY MELODIUS STANDS OUT

Unique Selling Points

01 Physical Key Mirroring

The only piano that physically activates keys at a remote location , not a screen, not an app, but a real tactile response.

02 Emotional Presence

Families, lovers, friends , feel as if the person is sitting right beside , playing. Music becomes a hug across the miles.

03 Near-Zero Latency

MQTT over WiFi delivers responses in milliseconds. The sync is so tight, latency is barely perceptible during natural play.

04 No App Required

Turn on, connect once, and play instantly. Inclusive for most ages and tech comfort levels.

05 Music Learning Tool

Enables real-time, physical teaching , empowering not only professionals but anyone with piano skills to share and teach, effortlessly

06 Beautifully Designed

Elegant, compact form factor inspired by the simplicity of a gift. Sits beautifully in any living room, bedroom, or studio.

WHO IS IT FOR?

Target Users & Scenarios

Distant Families

A grandparent plays a lullaby in Manchester while their grandchild hears and feels every key press in Paris, live.

Couples Apart

A couple in a long-distance relationship shares a nightly duet. One plays, the other feels it. Music as intimacy.

Friends & Collaborators

Two friends who grew up playing piano together, now in different cities, jam every Sunday morning as they always did.

Teachers & Students/Learners

A piano teacher or experienced piano player in London physically demonstrates fingering technique on the student's/learners piano in Liverpool , in real time.

REACH

28M+
Long-distance relationships globally (Lovedove.com) U.S.A

1 in 5
Families separated across borders

40M+
Piano players Worldwide (Qora.com)

TECHNOLOGY

How the MelodiUs Twin Piano Works

1

Wi-Fi Connected

User connects Wi-Fi through esp32 hotspot.

2

Key Pressed

User presses a device key at Location A

3

Local Feedback

The device plays the sound using I2S audio output, updates LED feedback

4

Event Transmission (MQTT)

At the same time, the event is sent via MQTT to the other device.

5

Remote Response

The second device receives the message and reproduces the same action with light + sound.

MQTT Broker

```
▼ucfnlwa
  ▼0021
    ▼B
      pressed = 0
    ▼A
      pressed = 0
```

Technical Highlights

1. Real-time audio synthesis using sine wave generation
2. Multiple keys can be played simultaneously by mixing waveforms
3. MQTT enables near real-time synchronisation between devices
4. Use LED patterns to indicate system status

Prototyping Journey

01 Concept & Research

- Identified need for remote shared musical interaction
- Defined physical key mirroring as core experience
- Positioned product between digital instruments and IoT interaction

02 Basic Interaction

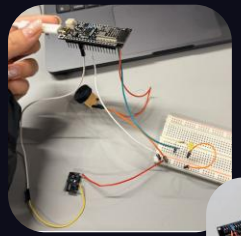
- Built first working prototype with key input + sound + LED
- Established basic interaction loop
- Validated physical feedback and responsiveness

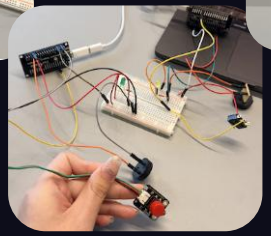
03 Connected System

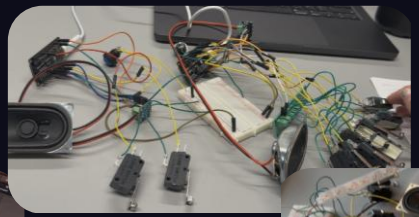
- Enabled real-time communication using MQTT
- Designed lightweight event-based protocol (D/U messages)
- Achieved low-latency key mirroring between devices
- Validated system responsiveness over Wi-Fi

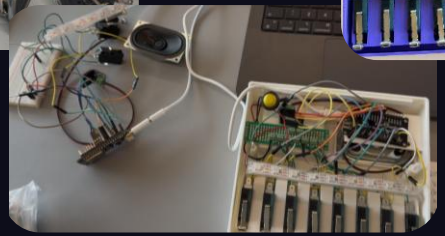
04 Integrated Product

- Scaled to full 8-key system per device
- Upgraded audio system (MAX98357A + speaker)
- Integrated LED feedback and enclosure design
- Delivered complete, usable product prototype







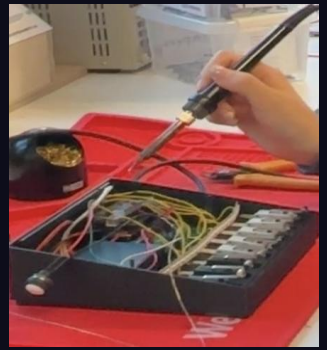


BUILD QUALITY

What We Built & How It Performs

✅ WHAT WORKS

- Real-time key mirroring with <500ms latency over local Wi-Fi
- Reliable MQTT communication across (100% message delivery in testing)
- Fully bidirectional interaction, both devices send and receive simultaneously
- Stable performance during extended play sessions (no disconnects observed)
- Responsive LED and audio feedback aligned with key input
- Integrated physical enclosure with refined finish and usability
- Simple setup , easy to use.



Current Limitations & Challenges

Internet Distance Latency

Over local Wi-Fi, latency is almost unnoticeable, enabling smooth real-time interaction. However, when connecting across long distances (e.g. London to China), network latency can introduce slight delays in synchronisation. This reflects a fundamental constraint of internet infrastructure, which we are actively working to optimise.

Mechanical Actuation Fidelity

Current servo motors enable functional key movement but do not yet replicate the full expressive dynamics of an acoustic piano. Variations in touch, such as forte and pianissimo, are limited in this prototype and will be enhanced in Phase 2.

Scale Limitation (8 Keys)

This prototype covers 8 keys , sufficient for basic melodies but falling short of a full key instrument. Scaling up requires additional expansion boards and motor channels.

Power & Portability

The current product is powered by basic batteries. For improved sustainability and longer usage, future versions will incorporate rechargeable battery systems. This will enable greater portability, allowing the product to be easily used across different rooms or in travel settings.

Wi-Fi Dependency

Both units require a stable WiFi connection. Weak or congested networks can introduce jitter. A dedicated pairing mode with QoS (Quality of Service) settings is planned.

Initial device Setup Experience for Non-Tech Users

The current onboarding process involves manual connection to the ESP32 Wi-Fi interface and configuration of network and MQTT settings. To enhance user experience, future iterations will introduce frictionless one-tap pairing via NFC or QR codes.

WHAT COMES NEXT

Future Phases & Development Roadmap

⭐ PHASE 2 - SOLENOID MOTOR UPGRADE

- Elaborate on the initial prototype of using solenoids with hall effect sensor to capture keynote pressed and mimic the same mechanism over the miles.
- Enhance look and feel of the piano design. Test various enclosures with smooth finishing and a clean aesthetic. S
- Solenoids deliver sharper, faster strikes with full velocity sensitivity, eliminating mechanical latency and creating an experience truly indistinguishable from a person playing in the room. This is the single most impactful upgrade for Phase 2.

P2 Solenoid Mechanical Sync

- Solenoid motors replacing push buttons
- 12 keys instrument scale
- Velocity-sensitive actuation
- Sub-5ms actuation response

P3 Network & Latency Refinement

- Cloud MQTT with QoS Level 2
- Adaptive buffering for long distances
- Dedicated Melodius pairing app
- End-to-end encrypted session keys

P4 Product & Design Polish

- Premium wood enclosure options
- NFC one-tap pairing
- Companion mobile app
- Gift set packaging for couples

COST & FUNDING

Unit Cost Breakdown & Launch Plan

£44–52
Cost per unit
hardware + mfg + dev

£100 / pair early bird
£200 / Full commercial price
Early Bird price
Early bird & full price

~45%
Profit margin
at early bird pricing

£30,000
Crowdfunding goal
Initial batch 200–300 units

UNIT COST BREAKDOWN

Hardware ESP32 · Keys · Speaker · Amp · LED · Battery · Misc	£20–26
Manufacturing 3D printed enclosure · Assembly materials	£5–7
Development Software development · Testing & iteration	£15
Additional Packaging · Miscellaneous costs	£4
TOTAL PER UNIT	**£44–52**

FUNDING & LAUNCH PLAN

Launch Target
June 2026
First production run ships

Crowdfunding Goal
£30,000
Covers solenoid upgrade + tooling investment

Early Bird Price
£100 / pair
Full commercial price £200 — 50% launch discount

Suggested Retail (single)
£79–99
~40–50% profit margin at full commercial pricing

We aim to finalise the product and ship by June 2026, depending on the crowdfunding we receive.

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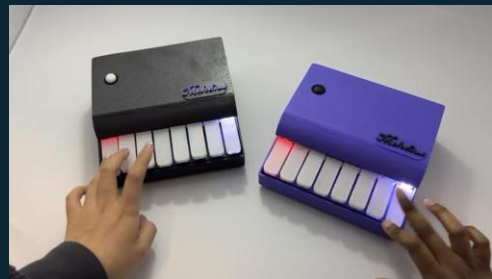
@melodiuspiano

Thank you - Q&A

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